GAZİ UNIVERSITY

FACULTY OF ENGINEERING
DEPARTMENT OF INDUSTRIAL ENGINEERING

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IE326-DATABASE MANAGEMENT SYSTEMS

WEEK 13: GROUPING DATA-MODIFY DATA

GROUPING DATA

GROUP BY, HAVING

GROUP BY

The `GROUP BY` clause divides the rows returned from the `SELECT` statement into groups.

For each group, you can apply an aggregate function such as `SUM()` to calculate the sum of items or `COUNT()` to get the number of items in the groups.

The following illustrates the basic syntax of the `GROUP BY` clause:

```
SELECT
    column_1,
    column_2,
    ...,
    aggregate_function(column_3)
FROM
    table_name
GROUP BY
    column_1,
    column_2,
    ...;
```

In this syntax:

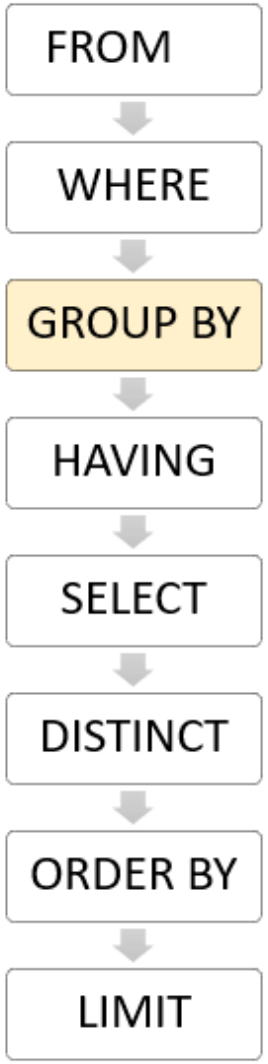
- First, select the columns that you want to group such as `column1` and `column2`, and column that you want to apply an aggregate function (`column3`).
- Second, list the columns that you want to group in the `GROUP BY` clause.

GROUP BY

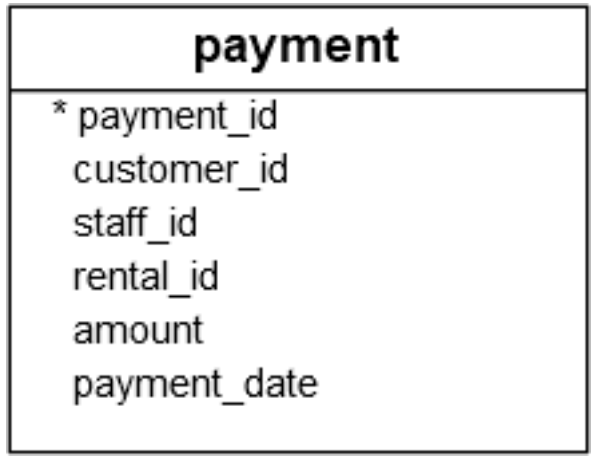
The `GROUP BY` clause divides the rows by the values in the columns specified in the `GROUP BY` clause and calculates a value for each group.

It's possible to use other clauses of the `SELECT` statement with the `GROUP BY` clause.

PostgreSQL evaluates the `GROUP BY` clause after the `FROM` and `WHERE` clauses and before the `HAVING`, `SELECT`, `DISTINCT`, `ORDER BY` and `LIMIT` clauses.



GROUP BY-EXAMPLE-DISTINCT



The following example uses the GROUP BY clause to retrieve the customer_id from the payment table:

```
SELECT customer_id FROM payment GROUP BY customer_id ORDER BY
customer_id;
```

```
customer_id
-------------
           1
           2
           3
           4
           5
           6
           7
           8
...
```

Each customer has one or more payments. The GROUP BY clause removes duplicate values in the customer_id column and returns distinct customer ids.

In this example, the GROUP BY clause works like the DISTINCT operator.

GROUP BY-EXAMPLE-SUM()

payment
* payment_id
customer_id
staff_id
rental_id
amount
payment_date

The GROUP BY clause is useful when used in conjunction with an aggregate function. The following query uses the GROUP BY clause to retrieve the total payment paid by each customer:

```
SELECT customer_id, SUM (amount)
FROM payment
GROUP BY customer_id
ORDER BY customer_id;
```

```
customer_id |  sum
------------+--------
          1 | 114.70
          2 | 123.74
          3 | 130.76
          4 |  81.78
          5 | 134.65
          6 |  84.75
          7 | 130.72
...
```

In this example, the GROUP BY clause groups the payments by the customer id. For each group, it calculates the total payment.

GROUP BY-EXAMPLE-SUM()

payment
* payment_id
customer_id
staff_id
rental_id
amount
payment_date

The following statement uses the ORDER BY clause with GROUP BY clause to sort the groups by total payments:

```
SELECT customer_id, SUM (amount) as TOTAL
FROM payment
GROUP BY customer_id
ORDER BY TOTAL DESC;
```

```
customer_id |  sum
------------+-------
        148 | 211.55
        526 | 208.58
        178 | 194.61
        137 | 191.62
        144 | 189.60
```

HAVING

The `HAVING` clause specifies a search condition for a group. The `HAVING` clause is often used with the `GROUP BY` clause to filter groups based on a specified condition.

The following statement illustrates the basic syntax of the `HAVING` clause:

```
SELECT
  column1,
  aggregate_function (column2)
FROM
  table_name
GROUP BY
  column1
HAVING
  condition;
```

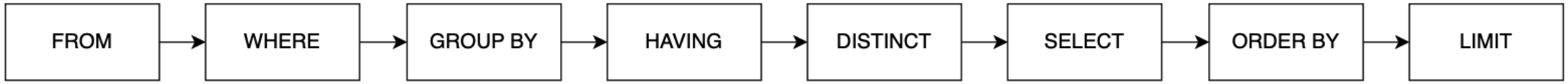
In this syntax:

- First, the `GROUP BY` clause groups rows into groups by the values in the `column1`.
- Then, the `HAVING` clause filters the groups based on the `condition`.

HAVING

Besides the `GROUP BY` clause, you can also include other clauses such as `JOIN` and `LIMIT` in the statement that uses the `HAVING` clause.

PostgreSQL evaluates the `HAVING` clause after the `FROM`, `WHERE`, `GROUP BY`, and before the `DISTINCT`, `SELECT`, `ORDER BY` and `LIMIT` clauses:



Because PostgreSQL evaluates the `HAVING` clause before the `SELECT` clause, you cannot use the column aliases in the `HAVING` clause.

This restriction arises from the fact that, at the point of `HAVING` clause evaluation, the column aliases specified in the `SELECT` clause are not yet available.

HAVING VS WHERE

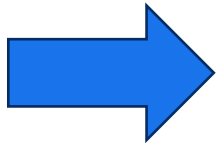
The `WHERE` clause filters the rows based on a specified condition whereas the `HAVING` clause filter groups of rows according to a specified condition.

In other words, you apply the condition in the `WHERE` clause to the rows while you apply the condition in the `HAVING` clause to the groups of rows.

HAVING EXAMPLE-SUM()

The following query uses the `GROUP BY` clause with the `SUM()` function to find the total payment of each customer:

```
SELECT
  customer_id,
  SUM (amount) amount
FROM
  payment
GROUP BY
  customer_id
ORDER BY
  amount DESC;
```



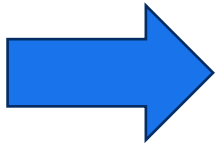
```
customer_id | amount
-------------+--------
          148 | 211.55
          526 | 208.58
          178 | 194.61
          137 | 191.62
...
```

payment
* payment_id
customer_id
staff_id
rental_id
amount
payment_date

HAVING EXAMPLE-SUM()

The following statement adds the `HAVING` clause to select the only customers who have been spending more than `200`:

```
SELECT
  customer_id,
  SUM (amount) amount
FROM
  payment
GROUP BY
  customer_id
HAVING
  SUM (amount) > 200
ORDER BY
  amount DESC;
```



```
customer_id | amount
-------------+--------
         148 | 211.55
         526 | 208.58
(2 rows)
```

payment
* payment_id
customer_id
staff_id
rental_id
amount
payment_date

HAVING EXAMPLE-COUNT

customer
* customer_id
store_id
first_name
last_name
email
address_id
activebool
create_date
last_update
active

The following query uses the GROUP BY clause to find the number of customers per store:

```
SELECT
  store_id,
  COUNT (customer_id)
FROM
  customer
GROUP BY
  store_id;
```

```
store_id | count
----------+-------
        1 |   326
        2 |   273
(2 rows)
```

The statement on the right adds the HAVING clause to select a store that has more than 300 customers:

```
SELECT
  store_id,
  COUNT (customer_id)
FROM
  customer
GROUP BY
  store_id
HAVING
  COUNT (customer_id) > 300;
```

```
store_id | count
----------+-------
        1 |   326
(1 row)
```

EXCERCISES- PREPARE YOUR LOCAL TABLE

```
-- Cleanup existing schema
DROP TABLE IF EXISTS order_details;
DROP TABLE IF EXISTS orders;
DROP TABLE IF EXISTS products;

-- 1. Create Tables
CREATE TABLE products (
    product_id SERIAL PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    category VARCHAR(50),
    unit_price NUMERIC(10, 2)
);

CREATE TABLE orders (
    order_id SERIAL PRIMARY KEY,
    customer_id INT NOT NULL,
    order_date DATE DEFAULT CURRENT_DATE,
    status VARCHAR(20)
);

CREATE TABLE order_details (
    detail_id SERIAL PRIMARY KEY,
    order_id INT REFERENCES orders(order_id),
    product_id INT REFERENCES products(product_id),
    quantity INT CHECK (quantity > 0)
);
```

```
-- 2. Insert Seed Data
INSERT INTO products (product_name, category, unit_price) VALUES
('Steel Sheet', 'Raw Material', 100.00),
('Copper Wire', 'Raw Material', 50.00),
('Gear Assembly', 'Component', 250.00),
('Motor', 'Component', 500.00),
('Welding Rod', 'Consumable', 15.00);

INSERT INTO orders (customer_id, order_date, status) VALUES
(501, '2024-01-10', 'Shipped'),
(502, '2024-01-12', 'Pending'),
(501, '2024-01-15', 'Shipped'),
(503, '2024-02-01', 'Shipped');

INSERT INTO order_details (order_id, product_id, quantity) VALUES
(1, 1, 10), (1, 3, 2), -- Order 1
(2, 4, 1),             -- Order 2
(3, 2, 20), (3, 3, 5), -- Order 3
(4, 5, 100);           -- Order 4
```

QUESTIONS

1- List each product category and the price of the most expensive item in that category. Note: Use MAX() function. Do some research on its usage.

2-Identify categories that have more than one product listed. Note: Use GROUP BY, HAVING AND COUNT()

3-Find the order_id and the total cost (quantity × unit_price) for every order, but only include orders where the total cost exceeds $1,000.

ANSWERS

1-SELECT category, MAX(unit_price) AS max_price
FROM products
GROUP BY category;

2-SELECT category, COUNT(*) AS product_count
FROM products
GROUP BY category
HAVING COUNT(*) > 1;

3- SELECT
    od.order_id,
    SUM(od.quantity * p.unit_price) AS total_order_value
FROM order_details od
JOIN products p ON od.product_id = p.product_id
GROUP BY od.order_id
HAVING SUM(od.quantity * p.unit_price) > 1000;

MODIFYING DATA

INSERT, UPDATE, DELETE, TRANSACTIONS

INSERT

The `INSERT` statement is used to add new rows of data to a table. You can insert a single row, multiple rows, or even data from another table.

---

1. Basic Syntax (Single Row)

To insert a row, specify the table name and the columns you want to fill, followed by the values.

SQL

```sql
INSERT INTO products (product_name, category, unit_price)
VALUES ('Induction Motor', 'Component', 750.00);
```

> **Tip:** If inserting values for **all** columns in the exact order they were created, the column names can be omitted. However, explicitly naming columns is safer for long-term code maintenance.

INSERT

INSERTING MULTIPLE ROWS

PostgreSQL allows inserting multiple rows in a single command, which is more efficient than running multiple separate queries.

SQL

```sql
INSERT INTO products (product_name, category, unit_price)
VALUES
    ('Valve Spring', 'Component', 12.50),
    ('Hydraulic Oil', 'Consumable', 45.00),
    ('Aluminum Plate', 'Raw Material', 85.00);
```

INSERTING DEFAULT VALUES

If a column has a `DEFAULT` value (like a `SERIAL` ID or a timestamp) or allows `NULL`, it can be omitted from the list.

SQL

```sql
-- order_date will automatically use the default value (e.g., CURRENT_DATE)
INSERT INTO orders (customer_id, status)
VALUES (505, 'Processing');
```

INSERT

RETURNING CLAUSE

The `RETURNING` clause is a PostgreSQL-specific feature that returns the data generated by the insert (such as an auto-incremented ID) without requiring a separate `SELECT` query.

SQL

```sql
INSERT INTO products (product_name, category, unit_price)
VALUES ('Industrial Sensor', 'Electronics', 120.00)
RETURNING product_id;
```

UPDATE

The `UPDATE` statement is used to modify existing records in a table. It is typically used with a `WHERE` clause to target specific rows.

To update a specific record, identify the table, the column to change, and the condition to locate the row.

SQL

```sql
UPDATE products
SET unit_price = 110.00
WHERE product_id = 1;
```

Tip: Always verify the `WHERE` clause before executing. If it is omitted, **all rows** in the table will be updated.

UPDATE

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To update a specific record, identify the table, the column to change, and the condition to locate the row.

SQL

```sql
UPDATE products
SET unit_price = 110.00
WHERE product_id = 1;
```

Tip: Always verify the `WHERE` clause before executing. If it is omitted, **all rows** in the table will be updated.

UPDATE

- Multiple columns can be updated at once by separating the column-value pairs with commas.

SQL

```sql
UPDATE products
SET category = 'Refined Material',
    unit_price = 55.00
WHERE product_id = 2;
```

- Updates can be performed using calculations based on existing values, such as applying a price increase.

SQL

```sql
-- Increase all prices in the 'Component' category by 10%
UPDATE products
SET unit_price = unit_price * 1.10
WHERE category = 'Component';
```

UPDATE

Similar to the `INSERT` statement, PostgreSQL allows the use of `RETURNING` to see the modified rows immediately.

SQL

```sql
UPDATE orders
SET status = 'Completed'
WHERE order_id = 102
RETURNING order_id, status;
```

UPDATE-Exercises

Q1. Basic Update: Write a query to change the status of order `101` to 'Delivered'.

SQL

```sql
UPDATE orders
SET status = 'Delivered'
WHERE order_id = 101;
```

Q2. Conditional Mass Update: Adjust the `unit_price` of all 'Raw Material' products by subtracting 5.00 from their current price.

SQL

```sql
UPDATE products
SET unit_price = unit_price - 5.00
WHERE category = 'Raw Material';
```

UPDATE-Exercises

Q3. Multi-column with Verification: Change the `product_name` to 'Heavy Motor' and `unit_price` to 600.00 for the product with `product_id = 4`. Return the updated row data.

SQL

```sql
UPDATE products
SET product_name = 'Heavy Motor',
    unit_price = 600.00
WHERE product_id = 4
RETURNING *;
```

DELETE

The `DELETE` statement is used to remove one or more rows from a table. Like `UPDATE`, it typically requires a `WHERE` clause to avoid unintended data loss.

To remove specific records, specify the table and the condition that identifies the rows to be deleted.

SQL

```sql
DELETE FROM products
WHERE product_id = 5;
```

Tip: If the `WHERE` clause is omitted, **all rows** in the table will be deleted, though the table structure itself will remain.

DELETE

- PostgreSQL allows you to return the data of the rows that were deleted. This is useful for archiving or verifying the operation.

```sql
DELETE FROM products
WHERE category = 'Consumable'
RETURNING *;
```

- PostgreSQL does not use a direct `DELETE JOIN` syntax like some other SQL dialects. Instead, it uses the `USING` clause to join other tables for filtering criteria.

```sql
-- Delete all order details for products categorized as 'Raw Material'
DELETE FROM order_details
USING products
WHERE order_details.product_id = products.product_id
  AND products.category = 'Raw Material';
```

DELETE

`CASCADE` is a property of a **Foreign Key constraint**, not a standalone command. When a table is created with `ON DELETE CASCADE`, deleting a row in the parent table automatically removes all related rows in the child table.

Example Setup:

SQL

```sql
CREATE TABLE order_details (
    detail_id SERIAL PRIMARY KEY,
    order_id INT REFERENCES orders(order_id) ON DELETE CASCADE,
    product_id INT REFERENCES products(product_id)
);
```

> **Tip:** With `ON DELETE CASCADE` enabled, running `DELETE FROM orders WHERE order_id = 101;` will automatically delete all rows in `order_details` that belong to order 101.

DELETE

Exercises

Q1. Basic Delete: Write a query to delete the product with `product_id = 3`.

```SQL
```

Q2. Filtered Delete: Remove all orders that have a status of 'Cancelled'.

```SQL
```

Q3. Delete with Using Delete all rows from the `order_details` table that are associated with `customer_id = 501`. (Note: `customer_id` is in the `orders` table).

```SQL
```

DELETE-DUPLICATE ROWS

In real-world databases, duplicate data can occur due to missing constraints or improper data imports. PostgreSQL provides several ways to identify and remove these duplicates while keeping one unique instance.

To find duplicates, use `GROUP BY` and `HAVING` on the columns that should be unique.

SQL

```sql
SELECT product_name, category, COUNT(*)
FROM products
GROUP BY product_name, category
HAVING COUNT(*) > 1;
```

DELETE-DUPLICATE ROWS-CTID

Every row in PostgreSQL has an internal, hidden column called `ctid` that represents the physical location of the row version. It is the simplest way to distinguish between two rows that have identical data.

SQL

```sql
DELETE FROM products a
USING products b
WHERE a.ctid < b.ctid
  AND a.product_name = b.product_name
  AND a.category = b.category;
```

Tip: In this query, we compare rows `a` and `b`. If they have the same data but different `ctid` values, the one with the smaller `ctid` is deleted, leaving the "newest" version.

TRANSACTIONS

A transaction is a single unit of work that contains one or more SQL statements. Transactions follow the ACID principle (Atomicity, Consistency, Isolation, Durability), ensuring that either all operations succeed or none of them are applied.

Basic Commands

- `BEGIN` : Starts a new transaction.
- `COMMIT` : Saves all changes made during the transaction to the database.
- `ROLLBACK` : Undoes all changes made since the `BEGIN` command.

TRANSACTIONS-STANDART WORKFLOW

A transaction is typically used when multiple related tables must stay in sync. For example, when a product is ordered, you must create an order record and decrease the product stock simultaneously.

SQL

```sql
BEGIN;

-- Step 1: Create the order
INSERT INTO orders (customer_id, status)
VALUES (700, 'Shipped');

-- Step 2: Record the details
INSERT INTO order_details (order_id, product_id, quantity)
VALUES (105, 1, 5);

-- Step 3: Finalize the changes
COMMIT;
```

TRANSACTIONS-ROLLBACK

If an error occurs during any step of the transaction (e.g., a product is out of stock), you can cancel the entire process to prevent partial data entry.

SQL

```sql
BEGIN;

UPDATE products
SET unit_price = unit_price + 10
WHERE category = 'Component';

-- If you realize you made a mistake (e.g., wrong category)
ROLLBACK;
```

Tip: In many database tools and drivers, if a single statement within a `BEGIN/COMMIT` block fails, the entire transaction is automatically marked as aborted and must be rolled back.

TRANSACTIONS-SAVEPOINTS

Savepoints allow you to roll back part of a transaction instead of the whole thing.

SQL

```sql
BEGIN;

INSERT INTO products (product_name, category) VALUES ('New Tool', 'Equipment');
SAVEPOINT sp1;

INSERT INTO products (product_name, category) VALUES ('Experimental Tool', 'Test');
-- Something went wrong with the experimental tool
ROLLBACK TO SAVEPOINT sp1;

-- This will only commit 'New Tool'
COMMIT;
```

Exercises

Question 1: Inventory & Sales Analysis (Medium)

Write a query to find each **category** that has a total potential revenue (calculated as `quantity * unit_price`) of more than $1,000. Display the category name and the calculated total revenue.

Question 2: Database Cleanup & Maintenance (Medium)

A business decision was made to stop selling products in the 'Consumable' category.

1. First, delete all records from the `order_details` table that involve 'Consumable' products using a `USING` clause.
2. Then, delete the 'Consumable' category products from the `products` table.

Exercises

Question 3: Safe Transaction Processing (Difficult)

A new customer (`customer_id = 605`) wants to place an order for 5 'Motors'. Write a complete **transaction** block that performs the following steps:

1. **Insert** a new order for the customer with the status 'Pending'.
2. **Insert** the order detail for 5 'Motors' (Hint: use a subquery or the known `product_id` for 'Motor').
3. **Update** the `unit_price` of 'Motors' in the `products` table, increasing it by 10% due to the new demand.
4. Ensure the entire operation is saved only if all steps succeed.

ANSWERS

A1

SQL

```sql
SELECT
    p.category,
    SUM(od.quantity * p.unit_price) AS total_category_revenue
FROM products p
JOIN order_details od ON p.product_id = od.product_id
GROUP BY p.category
HAVING SUM(od.quantity * p.unit_price) > 1000;
```

A2

SQL

```sql
-- Step 1: Delete details using a join-like filter
DELETE FROM order_details
USING products
WHERE order_details.product_id = products.product_id
  AND products.category = 'Consumable';

-- Step 2: Delete the products
DELETE FROM products
WHERE category = 'Consumable';
```

A3

SQL

```sql
BEGIN;

-- 1. Create the order
INSERT INTO orders (customer_id, status)
VALUES (605, 'Pending');

-- 2. Create the detail
-- Note: Assuming order_id is 5 based on serial increment
INSERT INTO order_details (order_id, product_id, quantity)
SELECT 5, product_id, 5
FROM products
WHERE product_name = 'Motor';

-- 3. Update the price
UPDATE products
SET unit_price = unit_price * 1.10
WHERE product_name = 'Motor';

COMMIT;
```

THE END

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GOT ANY QUESTIONS LATER?

SEND ME AN EMAIL: DR.ERCAN.EZIN@GMAIL.COM